Follow-up to the meeting of 8 November

Z + jets in Summer22: the muon channel, other b/c taggers, pileup, and the plot fixes

Same data and simulation files as the 237-page deck, so every number can be compared directly.

8 November 2026

Summary: the seven requests and what we found

1. Muon channel looks different from electrons	One real problem found: the muon trigger SF was computed from the leading muon only. Fixed. Z→μμ data/MC is now 0.943 (was 0.966), the same as Z→ee (0.949). The pT($\ell\ell$) shape did not change and is the same in electrons, so it is not a muon effect.
2. Other b/c taggers, Loose/Medium/Tight	DeepJet and RobustParT added next to ParticleNet, each with its own BTV SFs. Similar purity at the same working point. b-tag: data and MC agree for all three. c-tag: the taggers differ by a few percent.
3. Loose tau ID veto	Postponed, as agreed.
4. ee and μμ pileup look different	Our 20 test files of each channel come from different runs with slightly different pileup. That explains the difference.
5. ee vs μμ MC colours	Fixed: both channels now use the same stack and legend order.
6. MET up to ~200 GeV	Done. MC is well above data between 40 and 120 GeV (known 2022 jet/MET issue).
7. Zoom into high-statistics bins	Done: zoomed copies of 10 plots in every region.

Analysis setup 1/2: data, simulation, event selection

Data

- 2022 runs C–D, EGamma and Muon datasets, 22Sep2023 reprocessing, NanoAODv12.
- Certified lumisections only (golden JSON Cert_Collisions2022_355100_362760).
- Test sample: 20 files per channel = 389 pb^{-1} (ee) and 742 pb^{-1} (μμ), measured with brilcalc (normtag_PHYSICS) on the lumisections actually read.

Simulation (Run3Summer22, NanoAODv12)

- Drell–Yan: DYto2L-2Jets, MadGraph5_aMC@NLO with FxFx merging (0, 1, 2 partons at NLO), m($\ell\ell$) > 50 GeV. Inclusive sample used for generator pT($\ell\ell$) < 100 GeV, pT($\ell\ell$)-binned samples above (100–200, 200–400, 400–600, > 600).
- Backgrounds: $t\bar{t} \to 2\ell 2\nu$ (POWHEG), WW, WZ, ZZ (Pythia8).
- Each sample scaled to cross section × luminosity / sum of generator weights. 20 files per sample.

Jet flavour in MC

- Leading jet's hadronFlavour: b (a B hadron in the jet), c (a C hadron, no B), light (neither).

Triggers

- Z→ee: HLT_Ele30_WPTight_Gsf. Z→μμ: HLT_IsoMu24.

Leptons

- Electrons: pT > 25 GeV, |η| < 2.4, cut-based Tight ID, barrel–endcap gap 1.442 < |η_SC| < 1.566 removed, impact parameter |dxy| < 0.05 (0.1) cm and |dz| < 0.1 (0.2) cm in the barrel (endcap).
- Muons: pT > 26 GeV, |η| < 2.4, Tight ID, PF isolation (ΔR = 0.4) < 0.15 (Tight).

Z + jet selection (applied in this order)

- Trigger fired; two leptons of the channel, leading lepton pT > 35 GeV; opposite charge; 71 < m($\ell\ell$) < 111 GeV.
- Missing transverse momentum (type-1 PUPPI MET) < 50 GeV, to remove $t\bar{t}$.
- At least one AK4 jet (PUPPI): pT > 30 GeV, |η| < 2.5, Tight-lepton-veto ID (with the NanoAODv12 fix), ΔR > 0.4 from the leptons. The leading jet is the one we study and tag.

Analysis setup 2/2: corrections, scale factors, b/c tagging

All corrections from the CMS jsonPOG files (/cvmfs/cms-griddata.cern.ch/cat/metadata), Summer22 versions.

Jets and MET

- Jet energy corrections Summer22_22Sep2023_V4 (data per run era and MC); jet energy resolution smearing Summer22_22Sep2023_JRV2 (MC).
- Type-1 PUPPI MET recomputed with the corrected jets.
- Jet veto map Summer22_23Sep2023_RunCD_V1: event dropped if a jet (pT > 15 GeV, tight ID, EM fraction < 0.9, no muon nearby) is in a vetoed region.

Leptons

- Electron energy scale (data) and smearing (MC), EGM.
- Muon momentum scale (data) and smearing (MC), MUO. In Run 3 this replaces the Rochester correction.

Event cleaning

- MET filters in data and MC: goodVertices, globalSuperTightHalo2016, EcalDeadCellTriggerPrimitive, BadPFMuon, BadPFMuonDz, hfNoisyHits, eeBadSc.

Event weights in MC

- Pileup: LUM puWeights.json.gz (2022 C–D golden profile, 69.2 mb).
- Electrons: reconstruction and Tight ID SFs (EGM 2022 Re-reco BCD).
- Muons: Tight ID SF (NUM_TightID_DEN_TrackerMuons) and Tight isolation SF (NUM_TightPFIso_DEN_TightID), MUO.
- Trigger: either lepton can fire, so SF = $[1 - (1-\varepsilon^{1}\text{data})(1-\varepsilon^{2}\text{data})] / [1 - (1-\varepsilon^{1}\text{MC})(1-\varepsilon^{2}\text{MC})]$ with the POG data and MC efficiencies of both leptons (EGM Ele30 Tight ID; MUO IsoMu24 for Tight ID + Tight isolation).
- Not applied: L1 prefiring weight (not needed in Run 3), DY Z pT reweighting (switched off).

b and c tagging (leading jet only)

- Taggers: ParticleNet, DeepJet, RobustParT. Official BTV Summer22 Loose/Medium/Tight working points.
- b-tag: B score > cut. c-tag: CvL > cut and CvB > cut. b-tag and c-tag are separate selections.
- MC: the tagged jet is weighted by the BTV SF for its true flavour (b-tag: _comb for b and c, _light for light; c-tag: _wc for c, _tnp for b, _light for light). Outside the SF range the SF at the edge is used.

Uncertainties and software

- Statistical only (data and MC). No systematic or SF uncertainties yet.
- coffea 0.7, BTVNanoCommissioning (Zb_Zc_ratio workflow), correctionlib.

How to read the numbers in these slides

Z selection	Trigger, two leptons, opposite charge, 71 < m($\ell\ell$) < 111 GeV. No jet or MET requirement.
Full selection	Z selection + MET < 50 GeV + at least one jet. All plots use this unless stated.
data/MC	Data events divided by the MC prediction (all corrections and weights). 1 = perfect agreement.
Shape	data/MC in one range divided by the overall data/MC: the normalisation is taken out, only the shape is left.
Slope	Straight-line fit of data/MC versus pT, quoted in % per 100 GeV.
Test version	Muon study only: a copy of the muon selection with ONE thing changed, filled in the same run to see how much it matters. Tests are for comparison only and are not used in the analysis.
Tag ratio	Fraction of events with a tagged jet in data divided by the same fraction in MC. It tests the tagging and its SFs, independent of the overall normalisation.
Purity	Share of real b (or c) jets among the tagged jets in the DY simulation.
Errors	Statistical only (data and MC). Only 20 files per sample, so small differences are not significant.

1 · Muon channel: the question and how we checked it

Advisor: the muon channel shows a small discrepancy compared with electrons. Check the trigger turn-on, the η cut, the SFs and corrections, Rochester; follow the POG; maybe pT > 26 GeV and |η| < 2.4 help.

ingredient	what the analysis does now (POG recommendation)	changed?
Trigger SF	SF from the data and MC efficiencies of both muons	YES: was leading muon only (wrong)
Muon pT cut	both muons pT > 26 GeV (start of the POG trigger SFs)	YES: second muon was > 25 GeV
Leading muon pT	> 35 GeV (on the trigger plateau)	no
η cut	|η| < 2.4	no (already there)
Momentum correction	MUO scale and smearing (Run 3 replacement of Rochester)	no (already applied)
ID and isolation	Tight ID, Tight PF isolation, with their POG SFs	no (already applied)
Pileup weights	official LUM weights	no (already applied)

How we measured the effect of each ingredient

In the same run we also filled 10 test versions of the muon selection. Each changes ONE ingredient, e.g. switches the pileup weights off, so we can see how much data/MC moves. The tests are only for comparison: the analysis and all plots use the default setup with every correction applied.

1 · The muon trigger SF was wrong; now fixed

HLT_IsoMu24 fires if EITHER muon fires. The event trigger efficiency is $1 - (1-\varepsilon_1)(1-\varepsilon_2)$, which is close to 1. Our code used the SF of the leading muon only, as if only that muon could fire the trigger.

Example: one muon at η = 0.5, pT = 45 GeV, efficiencies from the MUO file

	efficiency in data	in MC	SF = data / MC
old: leading muon only	0.931	0.950	0.980
new: either of two such muons	0.9952	0.9975	0.998

The old SF lowered every MC event by about 2%. That made Z→μμ look better than it really was.

With the correct SF: Z→μμ data/MC 0.966 → 0.943. Z→ee was already done this way and stays at 0.949. The goal is a correct prediction, not data/MC = 1: the remaining 5–6% is common to both channels (two pages on).

Second change: both muons pT > 26 GeV (was 25 for the second muon)

The POG trigger efficiencies start at 26 GeV, so a muon below that cannot be used in the formula. Effect on data/MC: +0.3%.

1 · Tests (not applied): how much does data/MC depend on each ingredient?

First row: the default used in the analysis. Other rows: the same run with one thing changed, for comparison only. Slope: straight-line fit of data/MC vs pT(μμ), full selection, % per 100 GeV.

version	data/MC, full sel.	change	data/MC, Z sel.	change	slope
DEFAULT (used everywhere)	**0.943**	**–**	**0.986**	**–**	**-0.9 ± 1.2**
old setup (237-page deck)	0.966	+2.3%	1.012	+2.6%	-0.8 ± 1.2
trigger SF of leading muon only	0.969	+2.8%	1.014	+2.8%	-1.0 ± 1.2
other POG trigger method: match lead muon	0.936	-0.8%	0.981	-0.5%	-0.3 ± 1.3
second muon pT > 25 instead of 26	0.941	-0.3%	0.985	-0.1%	-0.8 ± 1.2
leading muon pT > 26 instead of 35	0.943	+0.0%	0.983	-0.4%	-0.9 ± 1.2
|η| < 2.1 instead of 2.4	0.944	+0.0%	0.989	+0.2%	-0.8 ± 1.3
without muon scale/smearing	0.941	-0.2%	0.985	-0.1%	-1.0 ± 1.2
without ID + isolation SFs	0.905	-4.0%	0.946	-4.1%	-0.4 ± 1.2
without pileup weights	0.958	+1.5%	0.990	+0.4%	-0.6 ± 1.2
without trigger SF	0.938	-0.6%	0.980	-0.6%	-0.8 ± 1.2

Only the scale factors move the normalisation. None of the choices changes the shape.

Slopes go from -1.0 to -0.3; the versions share almost all events, so this spread is the real effect. The two POG trigger methods agree within 0.8%.

1 · After the fix, muons and electrons agree at every step

data/MC for each group of events. The first four rows add up to all Z events; the last row is the final selection.

events	what they are	Z→ee	Z→μμ
all Z events	Z selection, no jet or MET requirement	0.995 ± 0.005	0.986 ± 0.003
MET < 50, no jet	Z + nothing else (most Z events)	1.042 ± 0.006	1.030 ± 0.004
MET > 50, no jet	large MET, no selected jet	0.416 ± 0.013	0.414 ± 0.008
MET > 50, ≥ 1 jet	large MET with jets (includes $t\bar{t}$)	0.692 ± 0.027	0.679 ± 0.019
final selection	**MET < 50 and ≥ 1 jet: the analysis sample**	**0.949 ± 0.012**	**0.943 ± 0.009**

The two channels now behave the same way everywhere.

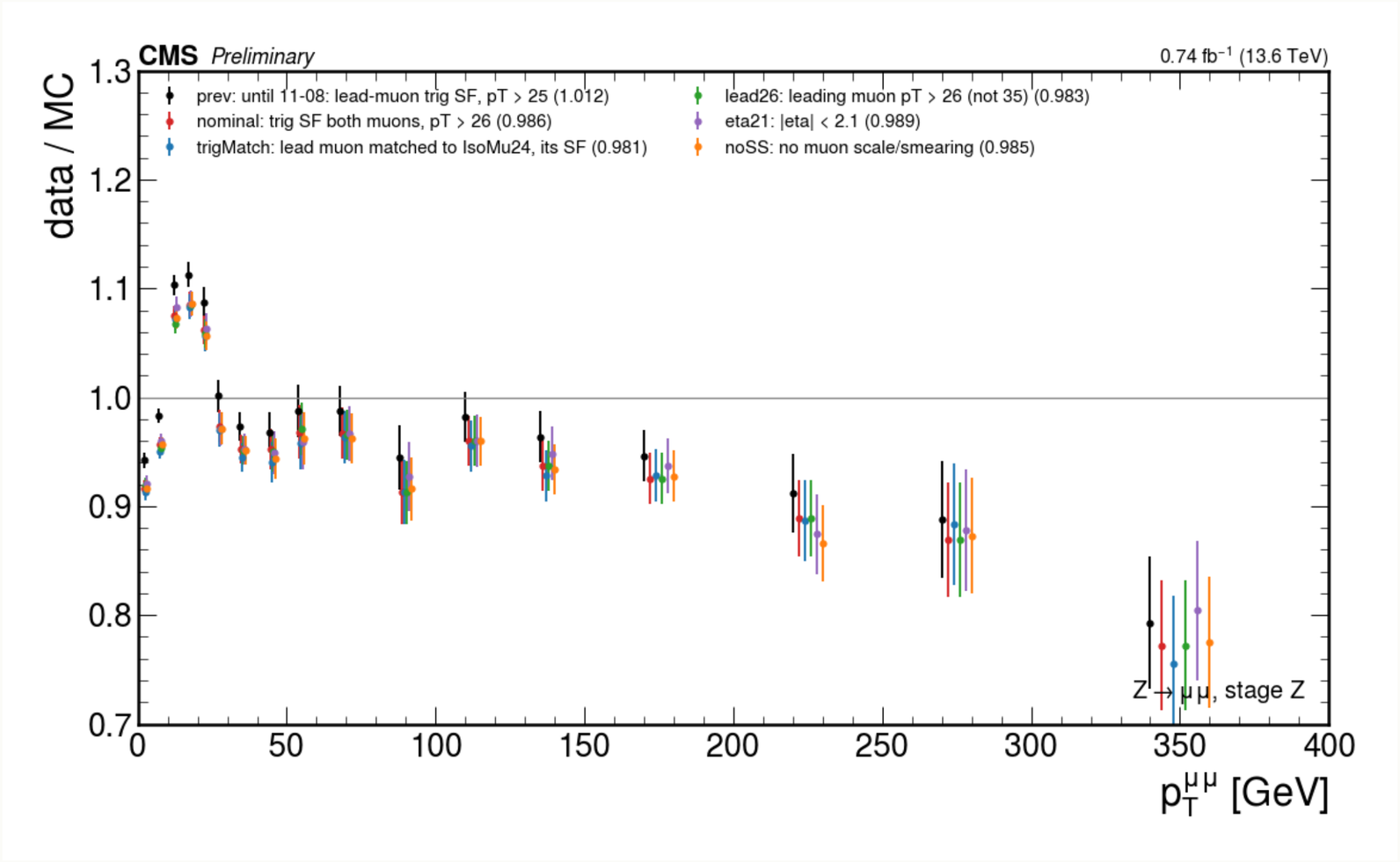
Before the fix, Z→μμ was about 2% higher than Z→ee because of the trigger SF.

The remaining 5–6% deficit is common to both channels, so it is not a lepton problem.

Events without jets have MORE data than MC (1.03–1.04); events with jets have less (0.94–0.95); events with MET > 50 GeV have far less (0.4–0.7). So the MC has too many events with jets and too much MET: jet modelling, not leptons.

1 · The muon tests do not change the pT(μμ) shape

data/MC versus pT(μμ) for six of the test versions, Z selection (before the jet and MET cuts). Black: old setup; red: default. The value in brackets is the overall data/MC of each version.



All versions follow the same curve; they only move up or down together.

So no muon setting is behind the pT trend. The same fall at high pT is seen in Z→ee (next pages).

1 · The pT(ℓℓ) trend: still there, unchanged, and the same in electrons

Shape of data/MC: data/MC in each pT(ℓℓ) range divided by the overall data/MC, full selection. 1.00 = same as average.

pT(ℓℓ) [GeV]	0–30	30–60	60–100	100–150	150–200	200–300	300–400	400–1000
Z→μμ, 237-page deck	1.020	0.975	1.015	1.041	0.996	0.982	0.805	1.011
Z→μμ, now	1.017	0.978	1.012	1.039	0.996	0.984	0.805	1.016
Z→ee (unchanged)	1.032	0.968	1.012	1.042	1.033	0.969	0.898	0.639
Z→μμ data events	16808	25207	13323	5068	1498	777	123	61

The visual trend is unchanged by the muon fixes, and Z→ee shows the same pattern.

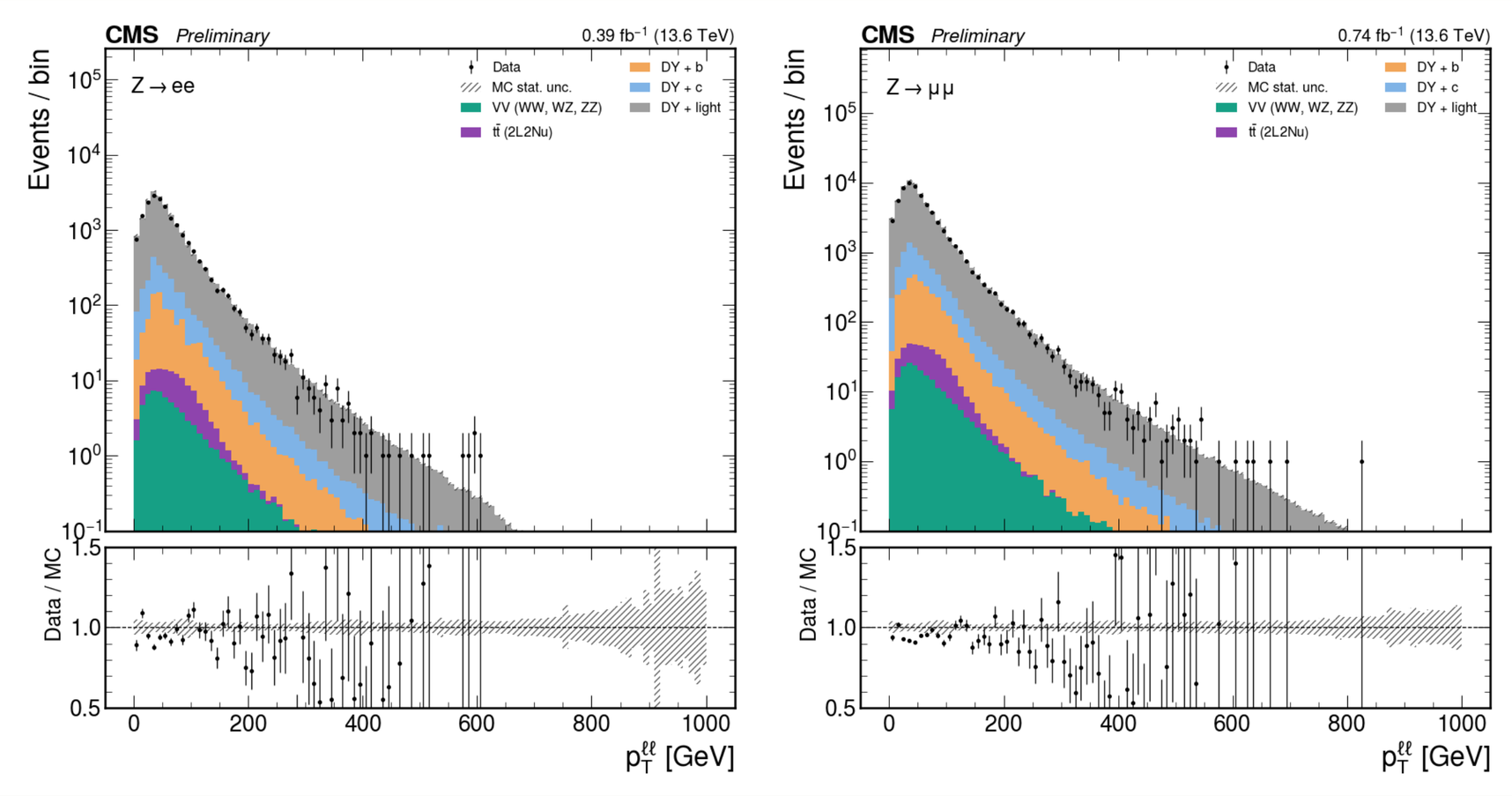
data/MC is about 2–3% low at 30–60 GeV, about 4% high at 100–150 GeV, and falls above 200 GeV (few events there).

It starts before any jet or MET cut, in both channels: the Drell–Yan simulation, not the leptons.

Slope at Z selection: Z→ee -2.1 ± 1.2, Z→μμ -3.5 ± 0.8 % per 100 GeV. Possible next steps: the full dataset (more events above 200 GeV), an official DY pT correction or another DY sample.

1 · The pT(ℓℓ) distribution as the advisor saw it (now with the fixes)

Full selection. Top: events per 10 GeV, MC stacked by jet flavour. Bottom: data/MC; hatched band = MC statistical uncertainty. Left Z→ee, right Z→μμ.



Same pattern in both channels: low at 30–60 GeV, high around 100–150 GeV, falling above 200 GeV.

Above ~300 GeV most bins have only a few events, so their scatter is mostly statistics.

1 · Where the earlier '+4.4% per 100 GeV' slope came from

The same straight-line fit of data/MC vs pT(ℓℓ) (10 GeV bins, 0–400 GeV, full selection) on three runs with the same files.

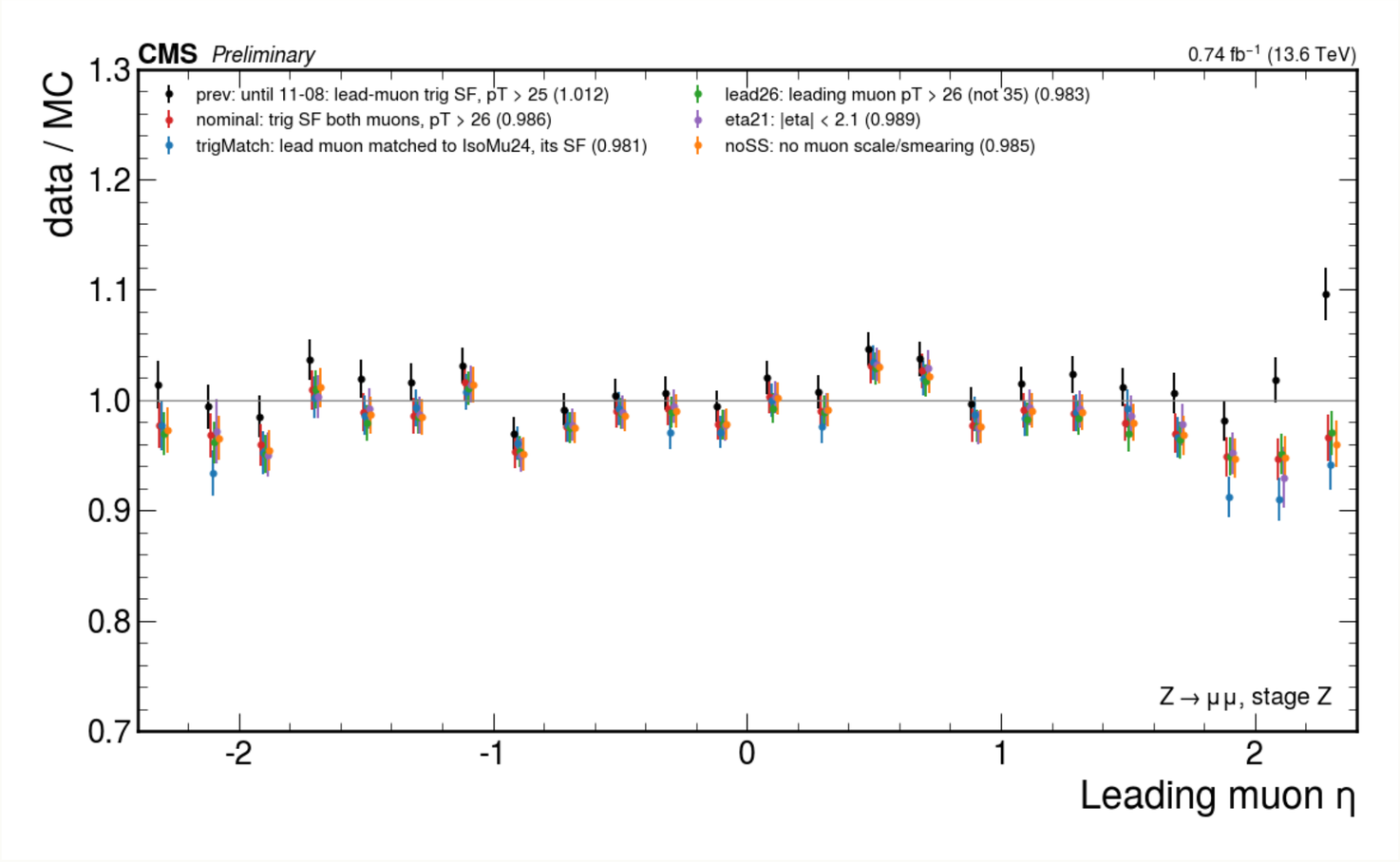
run	slope Z→ee	slope Z→μμ
29 Oct run: DY Z pT weights ON	+4.8 ± 2.0	+4.4 ± 1.3
237-page deck: weights OFF	-0.3 ± 1.9	-0.7 ± 1.2
now: weights OFF + muon fixes	-0.3 ± 1.9	-0.8 ± 1.2

The +4.4 came from our own Z pT weights, which over-corrected. They are off since the 237-page deck.

With the weights off, a straight line gives no significant slope, because the pattern goes up and then down (previous pages). The fit number alone hides it, so we look at the shape table.

1 · Still open: muons in the outer endcap (|η| > 2.0) are a few percent low

data/MC versus the leading-muon η, Z selection, for six of the test versions (same colours as before).



Default: 0.964 ± 0.010 at |η| 2.0–2.4, against 0.998 in the barrel.

The other POG trigger method gives 0.940 there, so it is not the formula. Cutting at |η| < 2.1 changes the overall data/MC by only +0.25%. Not understood yet; small effect on the total.

2 · Other taggers: DeepJet and RobustParT added next to ParticleNet

Advisor: try other b/c taggers with Loose, Medium and Tight working points.

tagger	working point	b-tag: B >	c-tag: CvL >	and CvB >
PNet	Loose	0.047	0.054	0.181
	Medium	0.245	0.16	0.306
	Tight	0.6734	0.492	0.259
DeepJet	Loose	0.0583	0.042	0.208
	Medium	0.3086	0.108	0.299
	Tight	0.7183	0.303	0.243
RobustParT	Loose	0.0849	0.039	0.068
	Medium	0.4319	0.117	0.13
	Tight	0.8482	0.36	0.095

What was done

The cut is applied to the leading jet. The working points are the official BTV Summer22 values. In MC the tagged jet gets the BTV SF of that tagger for its true flavour. Each tagger and working point is a separate set of plots: 18 tagged regions in total (3 taggers × L/M/T × b/c), all in hists_zbzc_run3_v2/Summer22/plots_<region>/.

2 · b-tagging: the three taggers compared

Purity: share of real b jets among the tagged DY jets (Z→μμ). Tagged: fraction of Z + jet events with a tagged leading jet (Z→μμ data). Tag ratio: tagged fraction in data / in MC; 1 = data and MC agree.

tagger	WP	purity	tagged	tag ratio Z→ee	tag ratio Z→μμ
PNet	L	22%	17.5%	0.998 ± 0.031	0.980 ± 0.022
	M	62%	5.8%	1.011 ± 0.055	0.969 ± 0.038
	T	88%	3.3%	1.058 ± 0.077	0.925 ± 0.047
DeepJet	L	20%	18.9%	0.976 ± 0.029	0.979 ± 0.022
	M	58%	6.0%	1.056 ± 0.056	0.996 ± 0.038
	T	87%	3.1%	1.050 ± 0.081	0.929 ± 0.048
RobustParT	L	23%	17.9%	1.013 ± 0.031	1.040 ± 0.024
	M	62%	5.7%	1.079 ± 0.061	1.006 ± 0.040
	T	89%	3.2%	1.034 ± 0.074	0.963 ± 0.050

Similar purity at the same working point, and data and MC tag the same fraction of jets for all three.

Largest difference from 1: RobustParT L Z→μμ, 1.040 ± 0.024 (1.7 standard deviations). SF uncertainties are not included yet.

2 · c-tagging: the three taggers compared

Purity: share of real c jets among the tagged DY jets (Z→μμ). Tagged: fraction of Z + jet events with a tagged leading jet (Z→μμ data). Tag ratio: tagged fraction in data / in MC; 1 = data and MC agree.

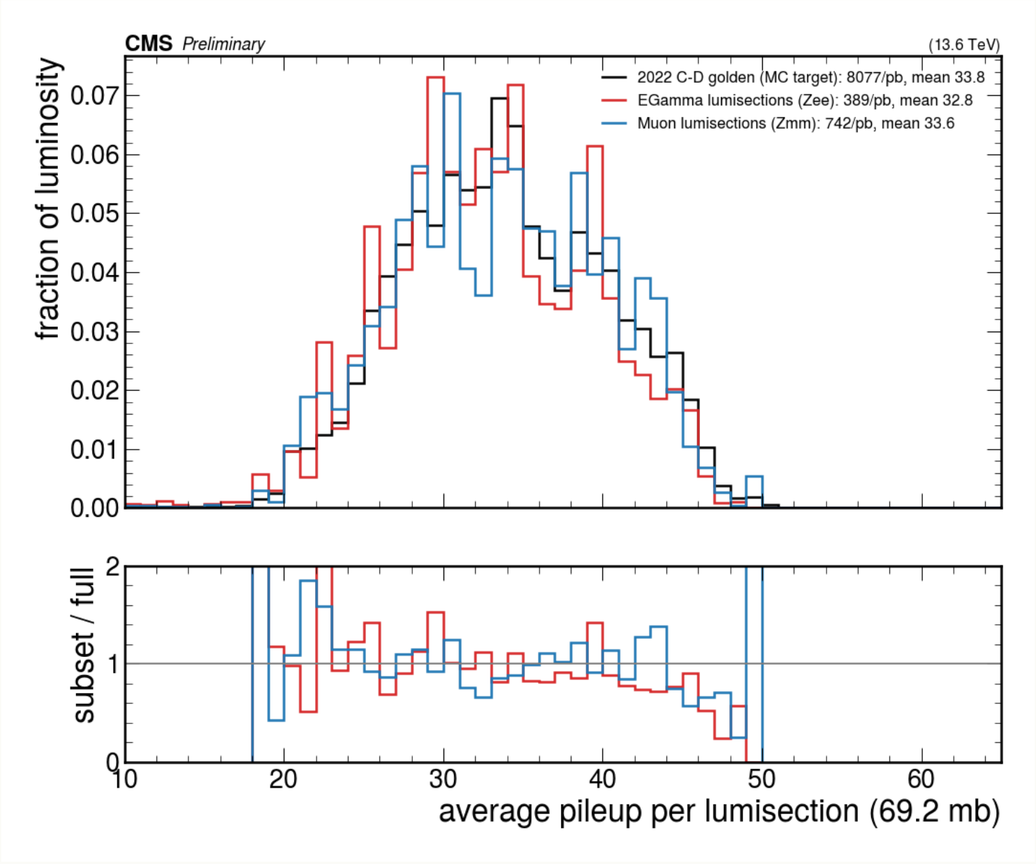
tagger	WP	purity	tagged	tag ratio Z→ee	tag ratio Z→μμ
PNet	L	9%	88.2%	0.981 ± 0.012	0.985 ± 0.009
	M	20%	28.1%	0.989 ± 0.023	0.985 ± 0.017
	T	51%	7.0%	0.965 ± 0.045	0.963 ± 0.035
DeepJet	L	9%	88.6%	1.015 ± 0.012	1.016 ± 0.009
	M	18%	27.6%	0.994 ± 0.024	1.025 ± 0.018
	T	48%	6.1%	1.088 ± 0.055	1.014 ± 0.038
RobustParT	L	9%	88.4%	1.014 ± 0.012	1.019 ± 0.009
	M	20%	27.5%	1.021 ± 0.024	1.045 ± 0.019
	T	54%	6.6%	1.099 ± 0.056	1.051 ± 0.039

Similar purity at the same working point, but the tag ratios differ by a few percent between taggers.

Average tag ratio: PNet 0.978, DeepJet 1.025, RobustParT 1.042. Largest: RobustParT M Z→μμ, 1.045 ± 0.019 (2.4 standard deviations, statistical only). SF uncertainties (a few % for c-tagging) are not included yet.

4 · Pileup: our ee and μμ test files come from different runs

Advisor: why do the ee and μμ pileup (number of vertices) plots look different? Below: how much luminosity was taken at each pileup value, for the lumisections our 20 test files cover (brilcalc), compared with all of 2022 C–D.



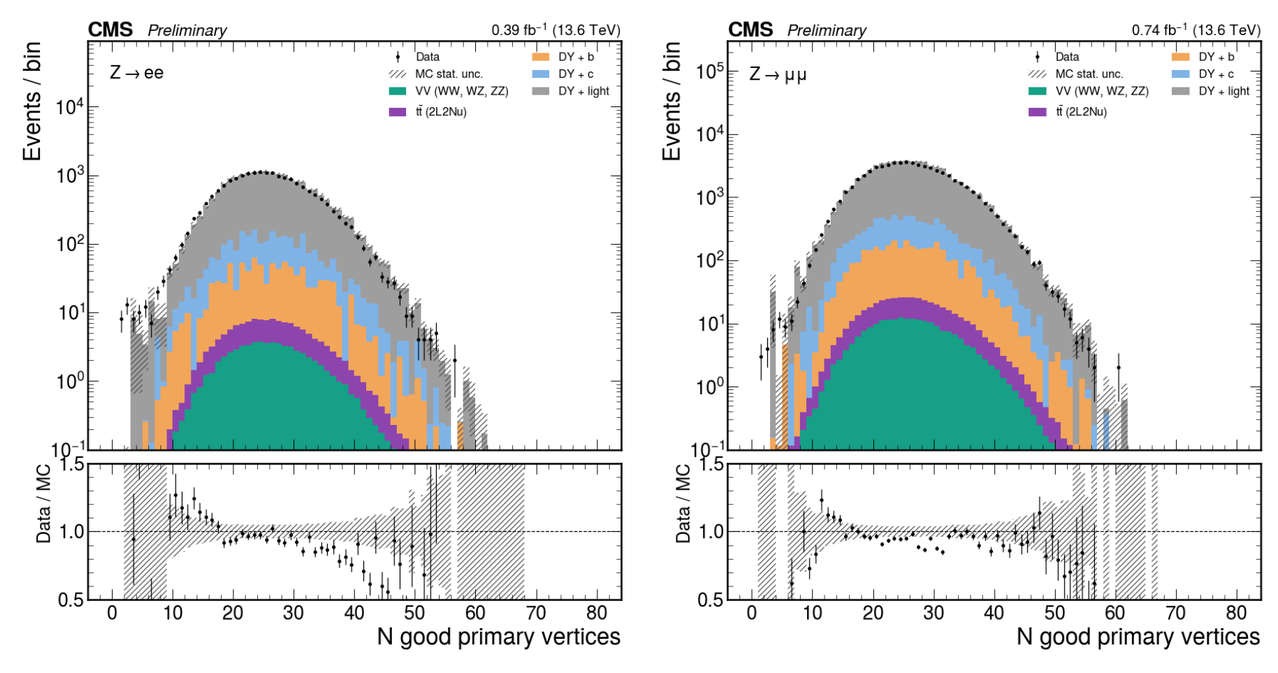
Mean pileup: EGamma files 32.8, Muon files 33.6, all of 2022 C–D 33.8.

The MC is weighted to all of 2022 C–D. The EGamma test files have less high-pileup data, so in Z→ee the data sit at fewer vertices than the MC.

4 · Pileup: the difference is explained by the runs in the test files

Check: if a channel's files have lower pileup, its data should have fewer vertices than its MC, by (pileup difference) × (vertices per pileup interaction). The MC gives 0.69 vertices per interaction.

	mean pileup	vs all 2022 C–D	expected vertex shift	observed (data − MC)
Z→ee (EGamma files)	32.84	-0.92	-0.64	-0.57
Z→μμ (Muon files)	33.56	-0.20	-0.14	-0.14

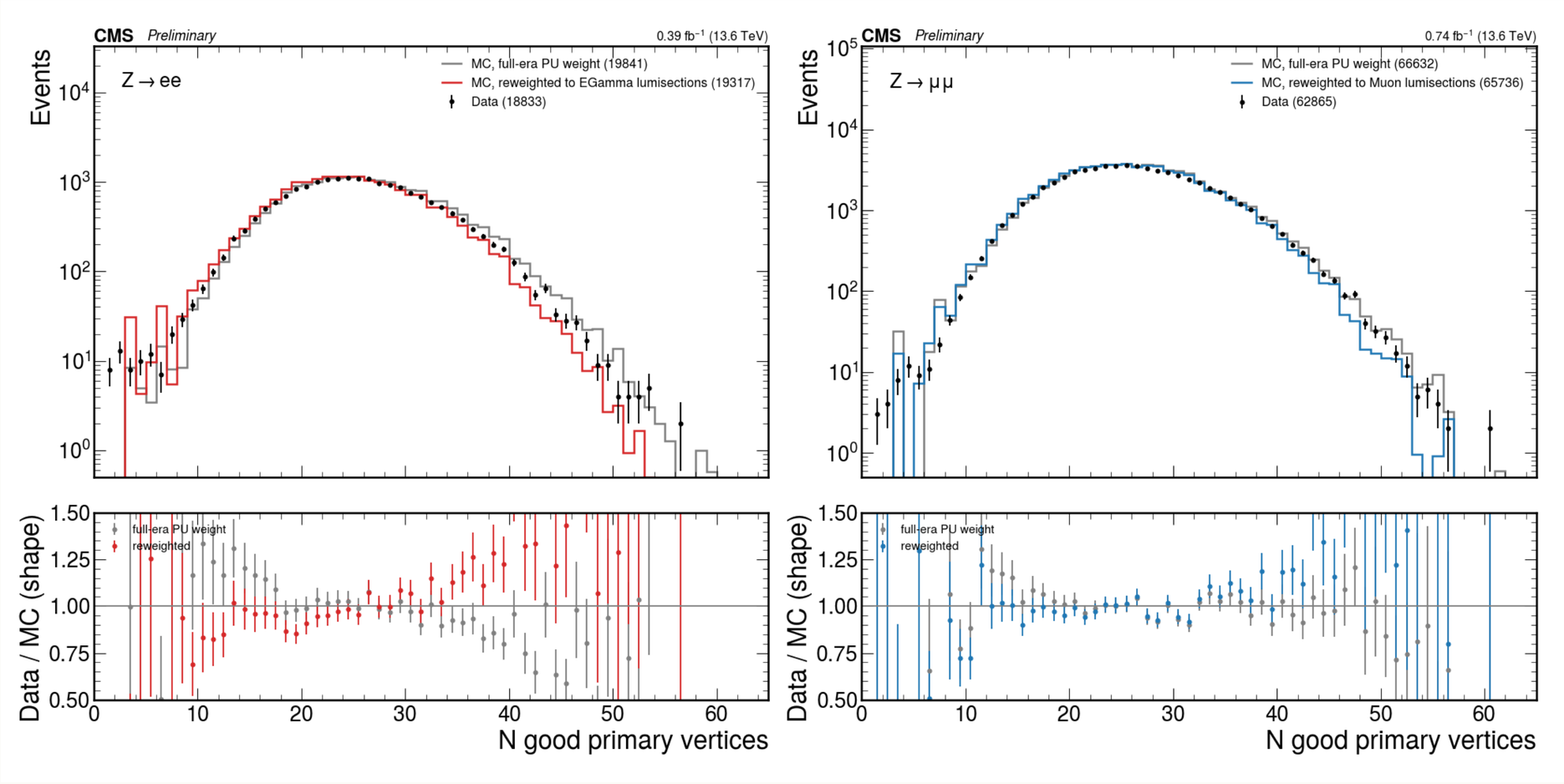


Expected and observed shifts agree in both channels. With the full dataset the difference should go away.

Plot: number of good primary vertices, full selection; left Z→ee, right Z→μμ.

4 · Pileup: what did not work

We also tried to reweight the MC bin by bin to each channel's own pileup profile (coloured) instead of the full 2022 C–D profile (grey). Bottom: data/MC shape.

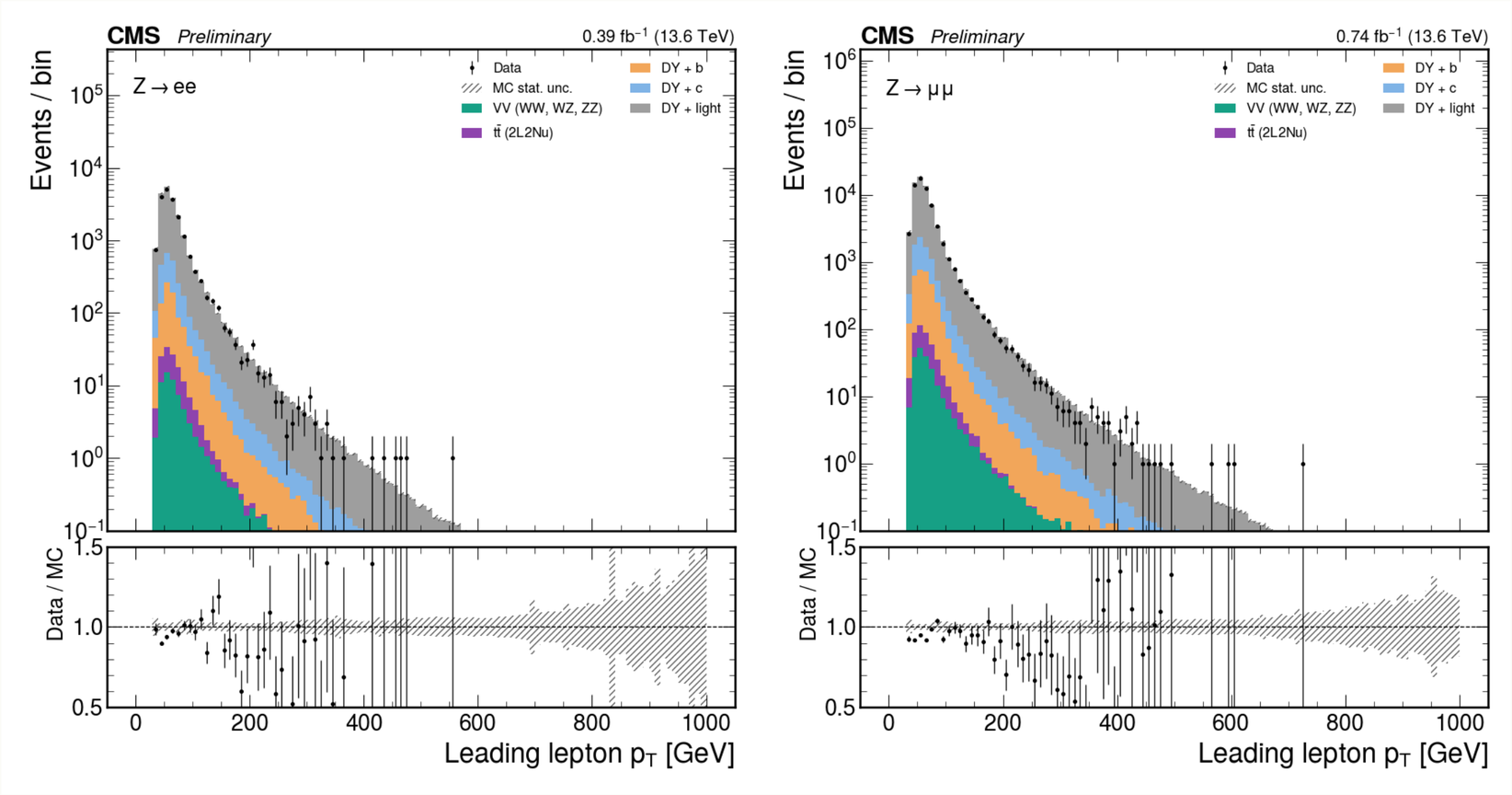


This over-corrects: Z→μμ, flat before, gets a slope. So it is not used.

Probably because a per-lumisection average ignores the bunch-to-bunch spread of pileup. The check on the previous page does not depend on that. The full-dataset run is the clean test.

5 · MC colours: the legend order is now the same in both channels

Advisor: in the Z→ee plots DY+b and DY+c were swapped in the legend. Example: leading-lepton pT, full selection.

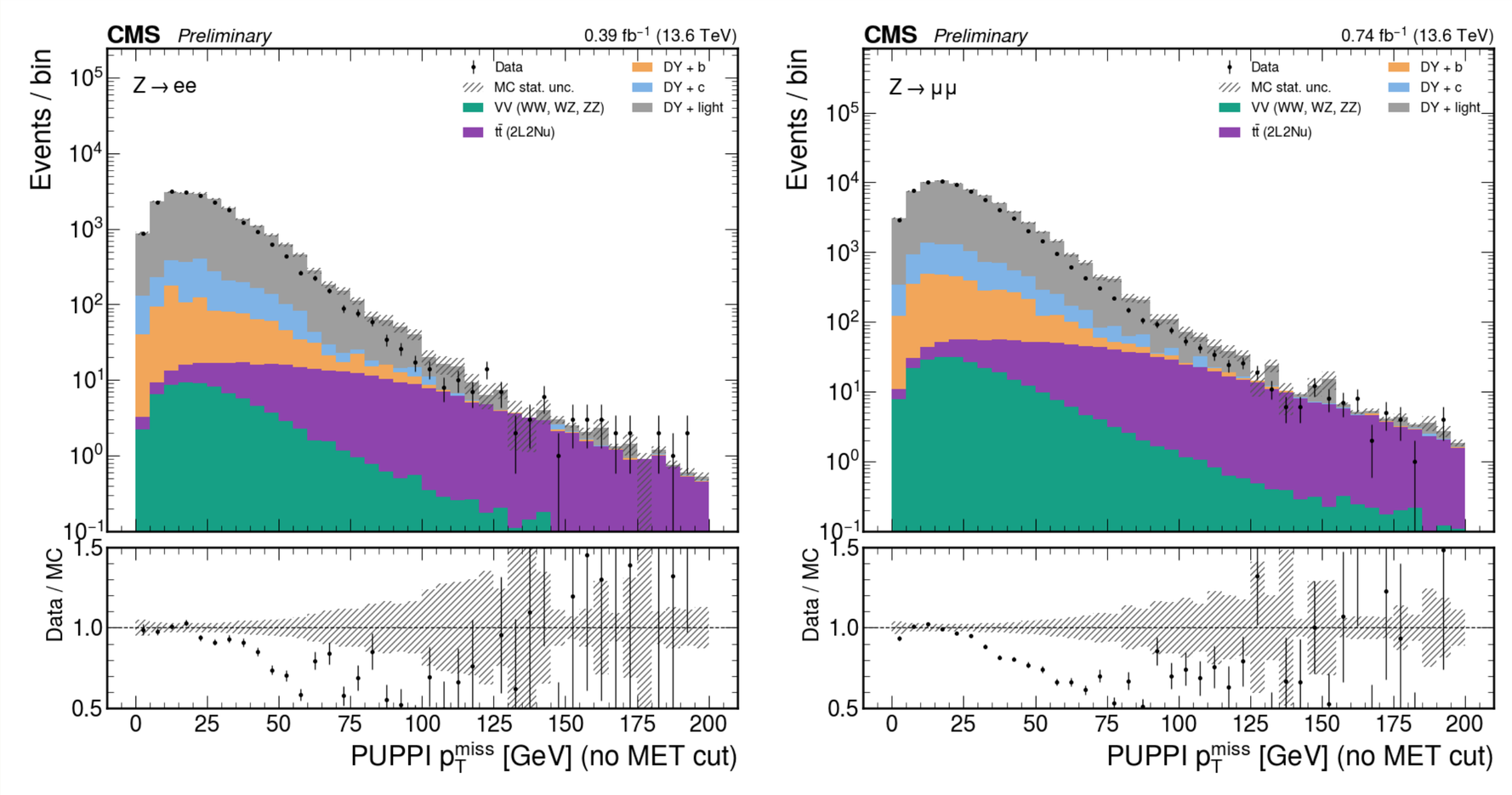


Fixed. Cause: each channel sorted the stack by its own yields, so the order changed when DY+b and DY+c had similar sizes.

Now one fixed order everywhere: DY+b, DY+c, DY+light, tt̄, VV. The colours were already the same. The 237-page deck's plots were remade too.

6 · MET up to 200 GeV, without the MET cut

Advisor: extend the MET plot to ~200 GeV. New plot: all cuts applied except MET < 50 GeV, so the region the analysis removes is visible. Available in every region.

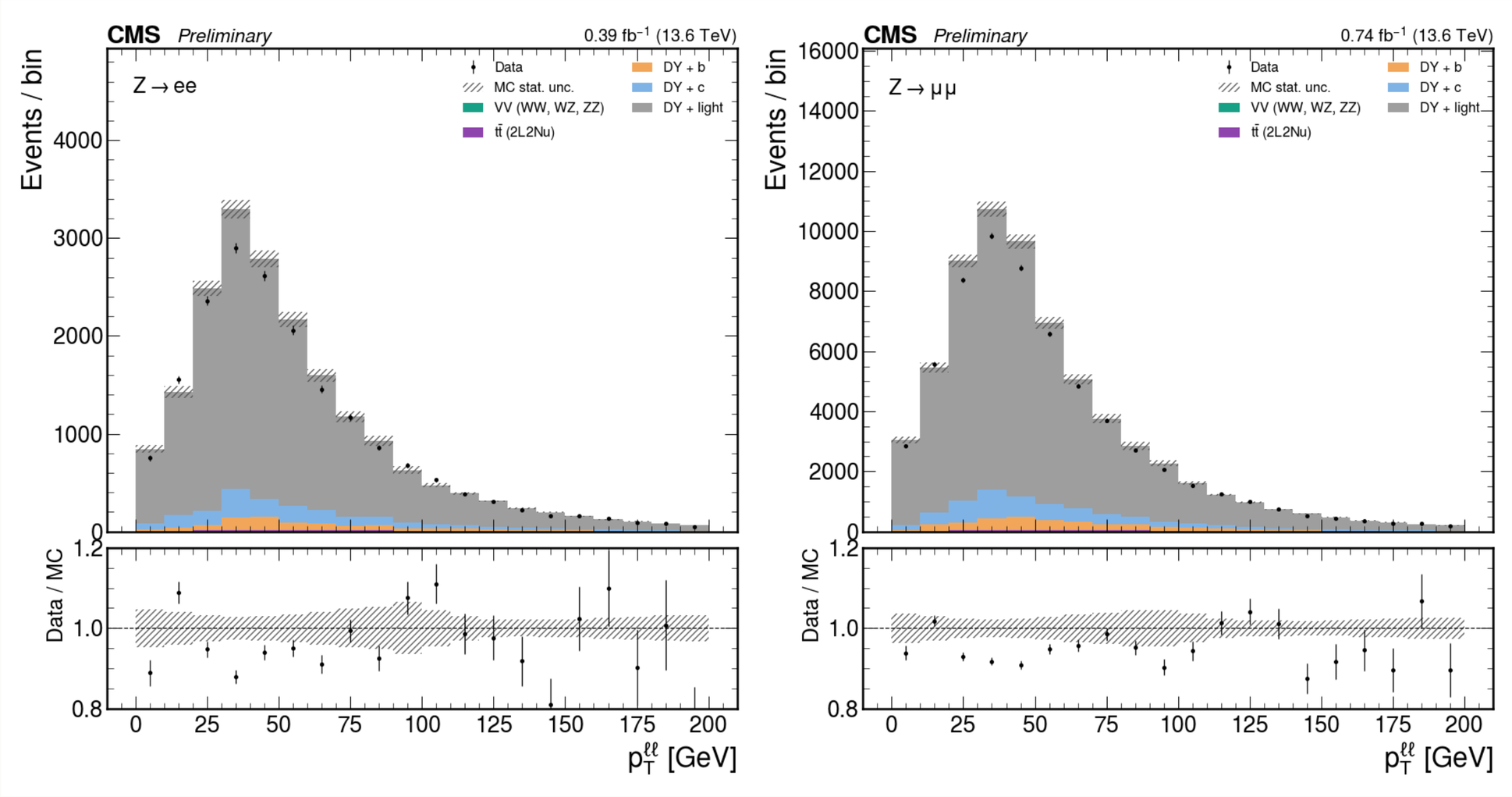


MC is well above data between about 40 and 120 GeV in both channels (data/MC down to ~0.6).

This is the 2022 jet/MET issue we traced in October (jets mismeasured in the MET). Above ~120 GeV $t\bar{t}$ dominates and agrees. Not fixed. The analysis cuts at 50 GeV, but the 40–50 GeV bins are already a bit low.

7 · Zoomed plots of the high-statistics region

Advisor: zoom into the high-statistics bins. Example: Z pT from 0 to 200 GeV, linear scale, data/MC from 0.8 to 1.2.



Done for 10 variables in every region (files *_zoom.png).

Z pT, both lepton pTs, leading and second jet pT, generator Z pT, MET (no MET cut), number of vertices, dijet mass and pT balance.

Checks on this run

Same files and lumisections as the 237-page deck.

Z→ee was not touched, and all 237 Z→ee histograms came out bit-for-bit identical. The old muon setup (one of the test versions) reproduces the old Z→μμ yields exactly (data 65259, MC 67588.4). So every difference in these slides comes from the intended changes.

Every sample read all of its files.

Two files had read timeouts on the first try and were rerun. One DY file cannot be read from the FNAL disk server at the moment, so the identical copy was read through the CMS redirector (same events). The job logs are now checked for the number of files read, not only for success.

Still open, and what we suggest next

Not understood or not fixed yet

- A 5–6% deficit after the jet and MET cuts, the same in both channels: MC has too many events with jets and with MET.
- The pT($\ell\ell$) pattern (low at 30–60 GeV, high at 100–150 GeV, falling above 200 GeV), the same in both channels.
- Muons at $|\eta| > 2.0$ are about 3% low before the jet and MET cuts.
- Loose tau ID veto (postponed). SF and systematic uncertainties. Only 20 files per sample so far.

To decide

- Keep the new muon setup (trigger SF from both muons, both muons pT > 26 GeV) as the default? We recommend yes: it is the POG prescription and makes the channels consistent.
- Which tagger and working point to use for b and for c.
- Then: run the full Summer22 dataset, add the SF uncertainties, and look at the pT($\ell\ell$) trend with an official DY correction.

Files (hists_zbzc_run3_v2/Summer22/)

- slides_zbzc_btagL_3taggers_Summer22.pdf: all kinematic plots, Loose b-tag, three taggers.
- plots_<region>/: 49 plots for each of the 19 regions. Tables: muvar/, pu/, tagger_summary.txt, channels_summary.txt.